

4. A laboratory technician tests **three** different food samples taken from a local restaurant by an Environmental Health Officer.

The samples were spread onto sterile agar plates which were then incubated at 37 °C.

After 3 days the agar plates were examined. The results are shown below.

It is assumed that each colony was formed from one bacterium.

Plate 1

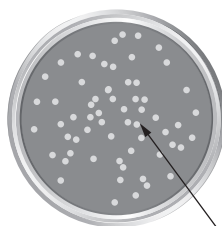


Plate 2

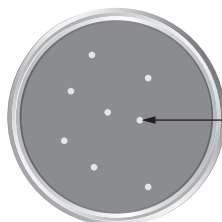
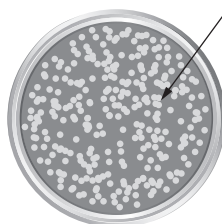


Plate 3



colonies of bacteria

- (a) (i) The technician concludes that the food that produced Plate 1 was stored in the restaurant at 15 °C, the food that produced Plate 2 was stored at 25 °C, and the food that produced Plate 3 was stored at 5 °C. Explain whether you agree. [3]

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- (ii) State why it is not possible to make a conclusion on the reproducibility of this analysis. [1]

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(b) Agar plates can also be used in a school laboratory to investigate bacterial growth.

(i) State why the agar plate needs to be sterile. [1]

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(ii) State what should be done to the inoculating loop before it is used to transfer the food sample to the agar plate. [1]

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(iii) Give a reason why the students are only allowed to incubate their plates at 25 °C. [1]

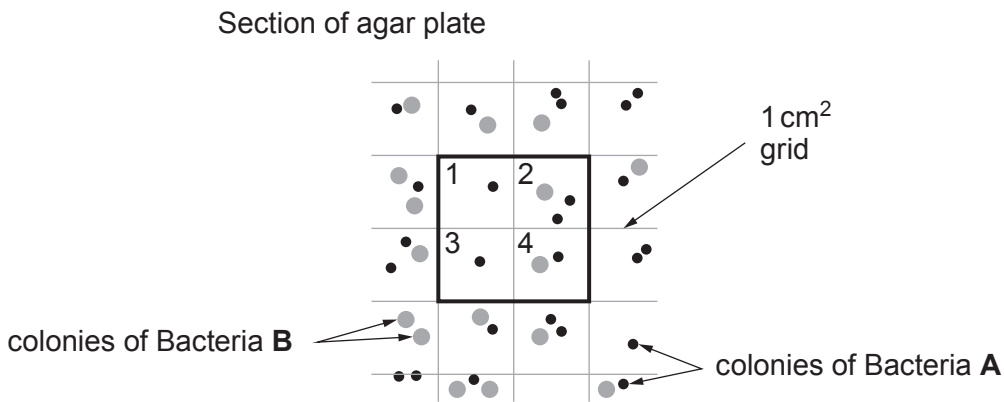
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(iv) State why it is important the lid remains on the agar plate during incubation. [1]

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- (c) Two types of bacteria, **A** and **B**, were found on one of the plates. The plate has an area of 52 cm². Each square in the grid is 1 cm².



The results for four squares numbered **1** to **4** are shown in the table below. The students have completed the table for Bacteria **A**.

Grid number	Number of bacteria in 1 cm ²	
	Bacteria A	Bacteria B
1	1	0
2	2	1
3	1	0
4	1	
Mean colonies per cm ²	1.25	
Mean colonies per 52 cm² plate	65	26
Total number of colonies in squares 1–4	5	

- (i) Complete the table for Bacteria **B**. [3]
- (ii) State how the results would be improved by counting the number of bacteria in 12 squares. [1]

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